An Exact Threshold and Transient Atlas for the Monod Chemostat

Route-A source-local certificate HCS-C254

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Abstract

For the one-species constant-yield Monod chemostat we expose an exact total- nutrient coordinate and use it to classify washout, equality and survival for all positive parameters. We also derive a separated transient on the invariant nutrient leaf and prove that every recurrent state is an equilibrium. This is an idealized source ODE, not biological calibration or an arithmetic determinant.

1 Frozen model and exact reduction

Let $D, S_{\rm in}, \mu_m, K, Y > 0$ and

$$\dot{S} = D(S_{\rm in} - S) - \frac{\mu(S)X}{Y}, \qquad \dot{X} = (\mu(S) - D)X, \qquad \mu(S) = \frac{\mu_m S}{K+S}. \tag{1}$$

The nonnegative quadrant is invariant. With $x = X/Y$ and $Q = S + x$,

$$\dot{Q} = D(S_{\rm in} - Q), \qquad Q(t) = S_{\rm in} + (Q_0 - S_{\rm in})e^{-Dt}. \tag{2}$$

Thus every solution is global and bounded.

2 Threshold theorem

Put $\Delta = \mu(S_{\rm in}) - D$.

Theorem 1. *If $\Delta < 0$, every state converges to $E_0 = (S_{\rm in}, 0)$. The same holds at $\Delta = 0$, with a nonhyperbolic biomass direction. If $\Delta > 0$, then*

$$S_* = \frac{DK}{\mu_m - D} < S_{\rm in}, \qquad X_* = Y(S_{\rm in} - S_*), \tag{3}$$

and every state with $X_0 > 0$ converges to $E_+ = (S_*, X_*)$; the invariant face $X_0 = 0$ converges to E_0.

Proof. Equation (2) gives $Q \to S_{\rm in}$. Substitution $S = Q - x$ gives $\dot{x} = [\mu(Q-x) - D]x$. Strict monotonicity of μ, together with scalar comparison at $S_{\rm in} \pm \varepsilon$, leaves zero as the limiting root when $\Delta \leq 0$ and the unique root $S_{\rm in} - S_*$ when $\Delta > 0$. The face $x = 0$ is invariant and must be separated. ◻

For completeness, after any fixed $\varepsilon > 0$ one has $S_{\rm in} - \varepsilon < Q(t) < S_{\rm in} + \varepsilon$. Monotonicity of μ brackets x between the two scalar logistic-type equations obtained by replacing $Q(t)$ with these endpoints. Their nonnegative equilibria converge to the roots stated above as $\varepsilon \downarrow 0$; this closes both $\liminf x$ and $\limsup x$ and does not assume local stability. At washout the eigenvalues are $-D, \Delta$. At survival, the (Q, x) eigenvalues are

$$-D, \qquad -(S_{\rm in} - S_*)\mu'(S_*). \tag{4}$$

The equality face is the transcritical exchange.

3 Exact attracting-leaf transient

On $Q = S_{\rm in}$ let $A = \mu_m - D$ and $x_* = S_{\rm in} - S_*$. Then

$$\dot{x} = \frac{Ax(x_* - x)}{K + S_{\rm in} - x}. \tag{5}$$

For $x_* \neq 0$, partial fractions give

$$\frac{K + S_{\rm in}}{x_*} \log x - \frac{K + S_*}{x_*} \log |x_* - x| = At + C. \tag{6}$$

At equality $x_* = 0$ this is replaced by

$$\frac{K + S_{\rm in}}{x} + \log x = At + C, \qquad x(t) \sim \frac{1}{\mu'(S_{\rm in})t}. \tag{7}$$

4 Recurrence obstruction

Theorem 2. *Every recurrent state of (1) is an equilibrium; in particular, the positive chemostat has no nonconstant periodic orbit.*

Proof. If a trajectory were periodic, (2) would make its Q coordinate periodic. The only periodic solution of the scalar relaxation equation is $Q = S_{\rm in}$. Equation (5) then governs the entire trajectory, and a one-dimensional autonomous flow has no nonconstant periodic solution. $\square$

5 Boundaries and evidence

If $X_0 = 0$, biomass stays zero. At $D = 0$, Q is conserved; for positive biomass substrate tends to zero. With $\mu_m = 0$ the system is linear, and $S_{\rm in} = 0$ forces $Q \to 0$. The faces $K = 0$ and $Y = 0$ change the model and are not reached by division.

face	exact reduced law	status
$X_0 = 0$	$X(t) = 0$	invariant washout boundary
$D = 0$	$Q(t) = Q_0$	closed batch limit
$\mu_m = 0$	linear exponential system	no-growth boundary
$S_{\rm in} = 0$	$Q(t) = Q_0 e^{-Dt}$	origin attracting

The executable certificate contains eighteen exact rational rows and five boundary contracts. An independent checker closes 244 assertions; fresh SymPy closes 14 identities; replay is byte-exact and 28/28 hostile semantic mutations are rehashed before testing. These finite receipts test algebra and conventions, not the continuous-parameter convergence proof. Monod supplies the named response convention; the 1977 SIAM paper supplies continuous-culture context; and the Smith–Waltman monograph supplies an authoritative model reference. No literature-priority claim is made.

6 Route-A boundary

The arithmetic origin is `none`. The strict tuple is `(A0_FAIL,A1_FAIL,A2_FAIL,A3_FAIL,A4_FAIL)` and the verdict is `ROUTE_A_REJECTED`, with `route_b_invocation_allowed: false`, under `NO_BAD_EULER_OR_ROOT_NUMBER`. No target divisor, functional equation, Euler factor, root number, automorphy or Hilbert–Pólya operator is claimed.

References

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[3] H. L. Smith and P. Waltman, *The Theory of the Chemostat*, Cambridge University Press, 1995.